

FIELD THEORY
B.E. Electrical Engineering
Second Year – Second Semester
2023 Examination – Part I
Complete Semester-Style Solution
Compile-Safe LaTeX Code

Important Compile Note

This code is intentionally made safe for Overleaf and VS Code.

It does not use the `physics` package.

It does not use `tikz-3dplot`.

It does not use `tdplot_main_coords`.

Therefore it avoids the common errors:

```
Command \divergence already defined
```

and

```
I do not know the key /tikz/tdplot_main_coords.
```

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1 Question 1(a): Electric Field at Midpoint of Hypotenuse

A right isosceles triangle of side 1 m has charges +1 nC, +2 nC and -1 nC arranged on its vertices. Find the magnitude and direction of electric field intensity at point P , which is the midpoint of the hypotenuse connecting the +1 nC and -1 nC charges. Medium is air.

1.1 Answer-Building Plan

1. Choose a convenient coordinate system.
2. Put the right angled vertex at origin.
3. Put +1 nC and -1 nC at the ends of the hypotenuse.
4. Put +2 nC at the right angled vertex.
5. Find coordinates of the midpoint.
6. Use Coulomb's law for electric field.
7. Add the three vector fields.
8. Find magnitude and direction.

1.2 Coordinate Arrangement

Let the right isosceles triangle have vertices:

$$A(0, 1), \quad B(1, 0), \quad C(0, 0).$$

Let

$$q_A = +1 \text{ nC} = +1 \times 10^{-9} \text{ C}$$

at $A(0, 1)$.

Let

$$q_B = -1 \text{ nC} = -1 \times 10^{-9} \text{ C}$$

at $B(1, 0)$.

Let

$$q_C = +2 \text{ nC} = +2 \times 10^{-9} \text{ C}$$

at $C(0, 0)$.

The midpoint of hypotenuse AB is

$$P = \left(\frac{0+1}{2}, \frac{1+0}{2} \right) = (0.5, 0.5).$$

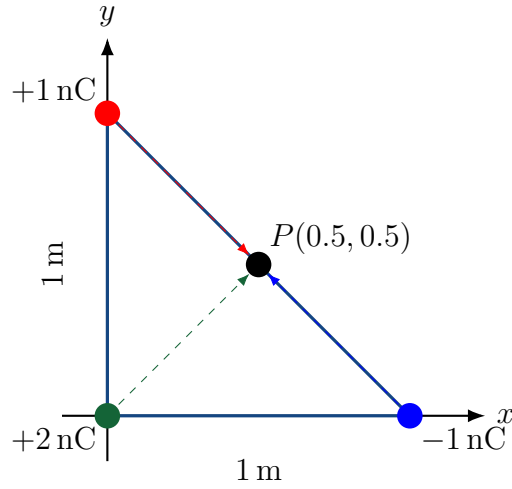


Figure 1: Coordinate arrangement for the right isosceles triangle.

1.3 Coulomb Field Formula

Electric field at point P due to a charge q is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} \mathbf{R}.$$

Here

$$\mathbf{R} = \mathbf{r}_P - \mathbf{r}_q.$$

For air,

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9.$$

1.4 Field Due to q_A

$$\mathbf{R}_A = (0.5 - 0)\hat{\mathbf{x}} + (0.5 - 1)\hat{\mathbf{y}}.$$

$$\mathbf{R}_A = 0.5\hat{\mathbf{x}} - 0.5\hat{\mathbf{y}}.$$

Magnitude:

$$R_A = \sqrt{(0.5)^2 + (-0.5)^2} = \sqrt{0.5} = \frac{1}{\sqrt{2}}.$$

Now,

$$\mathbf{E}_A = 9 \times 10^9 \frac{1 \times 10^{-9}}{(\sqrt{0.5})^3} (0.5\hat{\mathbf{x}} - 0.5\hat{\mathbf{y}}).$$

Since

$$(\sqrt{0.5})^3 = 0.5\sqrt{0.5},$$

we get

$$\mathbf{E}_A = 9\sqrt{2}\hat{\mathbf{x}} - 9\sqrt{2}\hat{\mathbf{y}}.$$

1.5 Field Due to q_B

$$\mathbf{R}_B = (0.5 - 1)\hat{\mathbf{x}} + (0.5 - 0)\hat{\mathbf{y}}.$$

$$\mathbf{R}_B = -0.5\hat{\mathbf{x}} + 0.5\hat{\mathbf{y}}.$$

$$R_B = \frac{1}{\sqrt{2}}.$$

Since charge at B is negative,

$$\mathbf{E}_B = 9 \times 10^9 \frac{-1 \times 10^{-9}}{(\sqrt{0.5})^3} (-0.5\hat{\mathbf{x}} + 0.5\hat{\mathbf{y}}).$$

Thus,

$$\mathbf{E}_B = 9\sqrt{2}\hat{\mathbf{x}} - 9\sqrt{2}\hat{\mathbf{y}}.$$

1.6 Field Due to q_C

$$\mathbf{R}_C = (0.5 - 0)\hat{\mathbf{x}} + (0.5 - 0)\hat{\mathbf{y}}.$$

$$\mathbf{R}_C = 0.5\hat{\mathbf{x}} + 0.5\hat{\mathbf{y}}.$$

$$R_C = \frac{1}{\sqrt{2}}.$$

Therefore,

$$\mathbf{E}_C = 9 \times 10^9 \frac{2 \times 10^{-9}}{(\sqrt{0.5})^3} (0.5\hat{\mathbf{x}} + 0.5\hat{\mathbf{y}}).$$

So,

$$\mathbf{E}_C = 18\sqrt{2}\hat{\mathbf{x}} + 18\sqrt{2}\hat{\mathbf{y}}.$$

1.7 Resultant Field

$$\mathbf{E} = \mathbf{E}_A + \mathbf{E}_B + \mathbf{E}_C.$$

$$\mathbf{E} = (9\sqrt{2} + 9\sqrt{2} + 18\sqrt{2})\hat{\mathbf{x}} + (-9\sqrt{2} - 9\sqrt{2} + 18\sqrt{2})\hat{\mathbf{y}}.$$

$$\mathbf{E} = 36\sqrt{2}\hat{\mathbf{x}} + 0\hat{\mathbf{y}}.$$

Thus,

$$|\mathbf{E}| = 36\sqrt{2}.$$

$$|\mathbf{E}| = 50.91 \text{ V/m}.$$

Since the y component is zero and the x component is positive, direction is along positive x -axis.

$$\mathbf{E} = 36\sqrt{2} \hat{\mathbf{x}} \text{ V/m}$$

$$|\mathbf{E}| = 50.91 \text{ V/m}$$

Direction: positive x -axis.

2 Question 1(b): Charge Enclosed by Unit Cylinder in Uniform Field

A cylinder of unit volume is placed in a uniform field with its axis parallel to the direction of electric field. Determine the total charge enclosed by the unit cylinder.

2.1 Concept

Gauss law in differential form is

$$\nabla \cdot \mathbf{D} = \rho_v.$$

If electric field is uniform, then electric flux density is also uniform in a homogeneous medium:

$$\mathbf{D} = \epsilon \mathbf{E}.$$

Uniform field means field has constant magnitude and constant direction.

Therefore,

$$\nabla \cdot \mathbf{D} = 0.$$

Hence,

$$\rho_v = 0.$$

Total charge enclosed is

$$Q = \iiint_V \rho_v dV.$$

Since $\rho_v = 0$,

$$Q = 0.$$

2.2 Integral Explanation

By Gauss law,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}.$$

The cylinder axis is parallel to the electric field.

Flux entering one circular face is equal to flux leaving the other circular face.

The curved surface contributes zero flux because electric field is parallel to the curved surface axis and perpendicular to the normal contribution needed.

Therefore net flux is zero.

So enclosed charge is zero.

$$Q_{\text{enc}} = 0 \text{ C}$$

3 Question 1(c): Electric Flux Lines and Equipotential Surfaces

Prove that electric flux line and equipotential are always normal to each other.

3.1 Definitions

An electric flux line is a line whose tangent at any point gives the direction of electric field $\mathbf{E}$.

An equipotential surface is a surface on which electric potential is constant.

So on an equipotential surface,

$$V = \text{constant}.$$

Therefore, for any small displacement $d\mathbf{l}$ along the equipotential surface,

$$dV = 0.$$

3.2 Proof

The electric field and potential are related by

$$\mathbf{E} = -\nabla V.$$

The differential change in potential is

$$dV = \nabla V \cdot d\mathbf{l}.$$

Using

$$\nabla V = -\mathbf{E},$$

we get

$$dV = -\mathbf{E} \cdot d\mathbf{l}.$$

But on an equipotential surface,

$$dV = 0.$$

Therefore,

$$-\mathbf{E} \cdot d\mathbf{l} = 0.$$

So,

$$\mathbf{E} \cdot d\mathbf{l} = 0.$$

Using dot product,

$$\mathbf{E} \cdot d\mathbf{l} = E dl \cos \theta.$$

Thus,

$$E dl \cos \theta = 0.$$

Since $E \neq 0$ and $dl \neq 0$,

$$\cos \theta = 0.$$

Therefore,

$$\theta = 90^\circ.$$

Hence electric field is normal to equipotential surface.

Since electric flux line is along electric field, electric flux line is also normal to equipotential surface.

Electric flux lines cut equipotential surfaces at right angles.

4 Question 2(a): Tangential Electric Field Boundary Condition

Derive the boundary condition for tangential component of electric field intensity on dielectric–dielectric interface.

4.1 Basic Idea

Consider two dielectric media separated by an interface.

Medium 1 has permittivity ϵ_1 .

Medium 2 has permittivity ϵ_2 .

Let the tangential components of electric field be

$$E_{1t}$$

and

$$E_{2t}.$$

We will prove

$$E_{1t} = E_{2t}$$

for electrostatic field.

4.2 Diagram

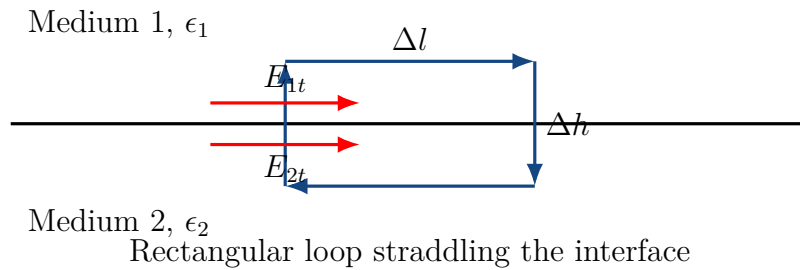


Figure 2: Loop used to derive tangential electric field boundary condition.

4.3 Derivation

For electrostatic field,

$$\nabla \times \mathbf{E} = 0.$$

Using Stokes' theorem,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0.$$

Apply this around a small rectangular loop crossing the boundary.

Contribution from upper horizontal side is

$$E_{1t}\Delta l.$$

Contribution from lower horizontal side is in opposite direction:

$$-E_{2t}\Delta l.$$

Contribution from vertical sides becomes zero as $\Delta h \rightarrow 0$.

Therefore,

$$E_{1t}\Delta l - E_{2t}\Delta l = 0.$$

$$(E_{1t} - E_{2t})\Delta l = 0.$$

Since $\Delta l \neq 0$,

$$\boxed{E_{1t} = E_{2t}}$$

4.4 Additional Relation for Electric Flux Density

Using

$$\mathbf{D} = \epsilon \mathbf{E},$$

we get

$$E_{1t} = \frac{D_{1t}}{\epsilon_1}$$

and

$$E_{2t} = \frac{D_{2t}}{\epsilon_2}.$$

Since

$$E_{1t} = E_{2t},$$

$$\boxed{\frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2}}$$

$$\boxed{E_{1t} = E_{2t}}$$

The tangential component of electric field intensity is continuous across a dielectric-dielectric interface.

5 Question 2(b): Volume Density of Energy in Electric Field

Starting from the work done in assembling the charges causing the electric field, derive the expression for volume density of energy in an electric field.

5.1 Energy in Assembling Charges

The work done in assembling a charge distribution becomes stored energy in the electric field.

For discrete charges,

$$W_e = \frac{1}{2} \sum q_i V_i.$$

The factor $\frac{1}{2}$ appears because pair interaction energy is counted twice in summation.

For continuous charge distribution,

$$W_e = \frac{1}{2} \iiint_V \rho_v V dv.$$

5.2 Using Gauss Law

Gauss law in differential form is

$$\nabla \cdot \mathbf{D} = \rho_v.$$

So,

$$W_e = \frac{1}{2} \iiint_V V(\nabla \cdot \mathbf{D}) dv.$$

Use vector identity

$$\nabla \cdot (V\mathbf{D}) = V(\nabla \cdot \mathbf{D}) + \mathbf{D} \cdot \nabla V.$$

Therefore,

$$V(\nabla \cdot \mathbf{D}) = \nabla \cdot (V\mathbf{D}) - \mathbf{D} \cdot \nabla V.$$

So,

$$W_e = \frac{1}{2} \iiint_V \nabla \cdot (V\mathbf{D}) dv - \frac{1}{2} \iiint_V \mathbf{D} \cdot \nabla V dv.$$

Using divergence theorem,

$$\iiint_V \nabla \cdot (V\mathbf{D}) dv = \oint_S V\mathbf{D} \cdot d\mathbf{S}.$$

If the integration region is extended to infinity, this surface term becomes zero for a finite charge distribution.

Hence,

$$W_e = -\frac{1}{2} \iiint_V \mathbf{D} \cdot \nabla V \, dv.$$

But,

$$\mathbf{E} = -\nabla V.$$

Therefore,

$$\nabla V = -\mathbf{E}.$$

So,

$$W_e = \frac{1}{2} \iiint_V \mathbf{D} \cdot \mathbf{E} \, dv.$$

Therefore energy density is

$$w_e = \frac{1}{2} \mathbf{D} \cdot \mathbf{E}$$

For a linear dielectric,

$$\mathbf{D} = \epsilon \mathbf{E}.$$

So,

$$w_e = \frac{1}{2} \epsilon E^2$$

Unit is

$$\text{J/m}^3.$$

$$w_e = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon E^2$$

6 Question 2(c): Validity of the Given Electric Field

Determine whether $\mathbf{E} = 3x\hat{\mathbf{x}} + 4y\hat{\mathbf{y}} - 5z\hat{\mathbf{z}}$ is a valid form of electric field or not.

6.1 Condition for Electrostatic Field

For a valid electrostatic field,

$$\nabla \times \mathbf{E} = 0.$$

This is because electrostatic field is conservative.

Given,

$$\mathbf{E} = 3x\hat{\mathbf{x}} + 4y\hat{\mathbf{y}} - 5z\hat{\mathbf{z}}.$$

So,

$$E_x = 3x, \quad E_y = 4y, \quad E_z = -5z.$$

Curl is

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x & 4y & -5z \end{vmatrix}.$$

Components are

$$(\nabla \times \mathbf{E})_x = \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = 0 - 0 = 0.$$

$$(\nabla \times \mathbf{E})_y = \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = 0 - 0 = 0.$$

$$(\nabla \times \mathbf{E})_z = \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0 - 0 = 0.$$

Therefore,

$$\nabla \times \mathbf{E} = 0.$$

So the field is a valid electrostatic field.

6.2 Additional Check

Divergence is

$$\nabla \cdot \mathbf{E} = \frac{\partial(3x)}{\partial x} + \frac{\partial(4y)}{\partial y} + \frac{\partial(-5z)}{\partial z}.$$

$$\nabla \cdot \mathbf{E} = 3 + 4 - 5 = 2.$$

If medium has permittivity ϵ ,

$$\rho_v = \nabla \cdot \mathbf{D} = \epsilon \nabla \cdot \mathbf{E} = 2\epsilon.$$

So this field can exist in a region with uniform volume charge density 2ϵ .

$$\nabla \times \mathbf{E} = 0$$

Therefore the given field is a valid electrostatic field.

7 Question 3(a): Field Just Off Conductor Surface

Justify or correct the statement giving reasons: “Electric field intensity just off the conductor surface is half of the electric field intensity on the conductor surface.”

7.1 Correct Physical Picture

For a conductor in electrostatic equilibrium,

$$\mathbf{E} = 0$$

inside the conductor.

Let surface charge density on conductor be

$$\rho_s.$$

The normal component of electric flux density just outside the conductor is

$$D_n = \rho_s.$$

Therefore,

$$\epsilon E_{\text{out}} = \rho_s.$$

So,

$$E_{\text{out}} = \frac{\rho_s}{\epsilon}$$

Inside conductor,

$$E_{\text{in}} = 0$$

7.2 Field Acting on Surface Charge

The field exactly acting on a surface charge is the average of field just outside and field just inside:

$$E_{\text{surface, acting}} = \frac{E_{\text{out}} + E_{\text{in}}}{2}.$$

Since

$$E_{\text{in}} = 0,$$

$$E_{\text{surface, acting}} = \frac{1}{2} E_{\text{out}}.$$

Therefore,

$$E_{\text{surface, acting}} = \frac{\rho_s}{2\epsilon}$$

7.3 Judgement of the Statement

The statement says that field just off the conductor surface is half of field on conductor surface.

This is not correct.

The correct statement is:

Field acting on the surface charge is half of the field just outside the conductor.

Equivalently,

$$E_{\text{out}} = 2E_{\text{surface, acting}}$$

The given statement is incorrectly worded. Correct form:

$$E_{\text{surface, acting}} = \frac{1}{2}E_{\text{out}}$$

8 Question 3(b): Mechanical Pressure on a Conductor Surface

Derive an expression for the mechanical pressure acting on a conductor surface.

8.1 Electric Field Near Conductor Surface

Let surface charge density on conductor be

$$\rho_s.$$

Electric field just outside the conductor is

$$E_{\text{out}} = \frac{\rho_s}{\epsilon}.$$

Electric field inside the conductor is

$$E_{\text{in}} = 0.$$

The average field acting on surface charge is

$$E_{\text{avg}} = \frac{E_{\text{out}} + E_{\text{in}}}{2} = \frac{E_{\text{out}}}{2}.$$

So,

$$E_{\text{avg}} = \frac{\rho_s}{2\epsilon}.$$

8.2 Force Per Unit Area

Force on charge is

$$F = qE.$$

For charge per unit area ρ_s , force per unit area is

$$p = \rho_s E_{\text{avg}}.$$

Substitute

$$E_{\text{avg}} = \frac{\rho_s}{2\epsilon}.$$

Then

$$p = \rho_s \frac{\rho_s}{2\epsilon}.$$

$$\boxed{p = \frac{\rho_s^2}{2\epsilon}}$$

Since

$$\rho_s = \epsilon E,$$

$$p = \frac{(\epsilon E)^2}{2\epsilon}.$$

Therefore,

$$p = \frac{1}{2}\epsilon E^2$$

This pressure acts normally outward on the conductor surface.

$$p = \frac{\rho_s^2}{2\epsilon} = \frac{1}{2}\epsilon E^2$$

9 Question 3(c): Impregnation of Porous Dielectric

Justify or correct the statement giving reasons: “Impregnation of porous solid dielectric should be done using a liquid dielectric whose relative permittivity is widely different from that of the solid dielectric.”

9.1 Physical Meaning

A porous solid dielectric contains voids or pores.

If these pores are filled with air, breakdown may occur easily because air has low dielectric strength.

Therefore, pores are filled with an insulating liquid.

This process is called impregnation.

9.2 Electric Field Sharing

At the interface of two dielectric materials, normal electric flux density is continuous if no free surface charge is present:

$$D_{1n} = D_{2n}.$$

Since

$$D = \epsilon E,$$

we get

$$\epsilon_1 E_1 = \epsilon_2 E_2.$$

Therefore,

$$\frac{E_1}{E_2} = \frac{\epsilon_2}{\epsilon_1}.$$

This means electric field is higher in the material having lower permittivity.

If relative permittivities of solid dielectric and liquid dielectric are widely different, electric stress becomes very non-uniform.

The low permittivity region gets excessive electric field.

This may cause dielectric breakdown.

9.3 Correct Statement

The relative permittivity of impregnating liquid should be close to the relative permittivity of the solid dielectric.

Then electric field distribution becomes more uniform.

So the given statement is wrong.

Correct statement:

Impregnating liquid should have relative permittivity close to that of the solid dielectric.

10 Question 4(a): Single Core Cable with Three Dielectric Media

In the case of a single core cable having three different dielectric media, explain under what condition the maximum electric field intensity will not occur just off the inner conductor surface. Derive necessary expressions.

10.1 Cable Geometry

Consider a single core cable.

Let the conductor radius be a .

Let the first dielectric extend from a to b with permittivity ϵ_1 .

Let the second dielectric extend from b to c with permittivity ϵ_2 .

Let the third dielectric extend from c to d with permittivity ϵ_3 .

Let charge per unit length on inner conductor be λ .

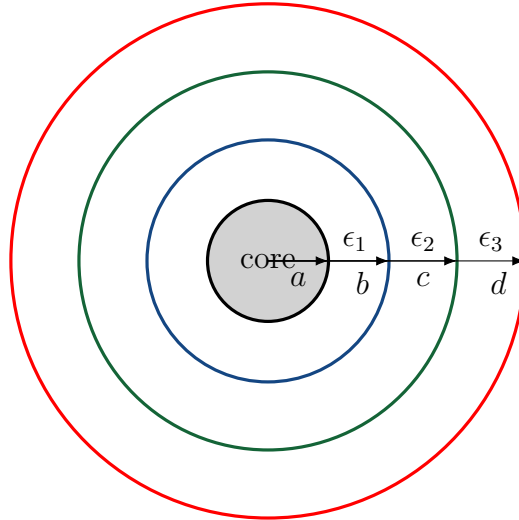


Figure 3: Single core cable with three dielectric layers.

10.2 Electric Field in Each Dielectric

By Gauss law, for a cylindrical Gaussian surface of radius r and length l ,

$$D(2\pi rl) = \lambda l.$$

Therefore,

$$D = \frac{\lambda}{2\pi r}.$$

Since

$$D = \epsilon E,$$

the electric field is

$$E = \frac{\lambda}{2\pi\epsilon r}.$$

Thus in the first dielectric,

$$E_1(r) = \frac{\lambda}{2\pi\epsilon_1 r} \quad a < r < b$$

In the second dielectric,

$$E_2(r) = \frac{\lambda}{2\pi\epsilon_2 r} \quad b < r < c$$

In the third dielectric,

$$E_3(r) = \frac{\lambda}{2\pi\epsilon_3 r} \quad c < r < d$$

10.3 Maximum Stress in Each Layer

Since $E \propto \frac{1}{r}$ in each layer, maximum stress in each layer occurs at the inner radius of that layer.

For first layer,

$$E_{1\max} = \frac{\lambda}{2\pi\epsilon_1 a}.$$

For second layer,

$$E_{2\max} = \frac{\lambda}{2\pi\epsilon_2 b}.$$

For third layer,

$$E_{3\max} = \frac{\lambda}{2\pi\epsilon_3 c}.$$

Usually maximum stress occurs at just outside the inner conductor, i.e. at $r = a$.

That happens if

$$E_{1\max} > E_{2\max}$$

and

$$E_{1\max} > E_{3\max}.$$

But maximum field will not occur just off the inner conductor if either

$$E_{2\max} > E_{1\max}$$

or

$$E_{3\max} > E_{1\max}.$$

That is,

$$\frac{\lambda}{2\pi\epsilon_2 b} > \frac{\lambda}{2\pi\epsilon_1 a}$$

or

$$\frac{\lambda}{2\pi\epsilon_3 c} > \frac{\lambda}{2\pi\epsilon_1 a}.$$

Canceling common terms,

$$\boxed{\epsilon_1 a > \epsilon_2 b}$$

or

$$\boxed{\epsilon_1 a > \epsilon_3 c}$$

So the maximum will not occur just outside the inner conductor if $\epsilon_1 a$ is not the least among

$$\epsilon_1 a, \quad \epsilon_2 b, \quad \epsilon_3 c.$$

10.4 Condition for Equal Maximum Stresses

For ideal grading, maximum stress in all three dielectrics should be equal:

$$E_{1\max} = E_{2\max} = E_{3\max}.$$

So,

$$\frac{\lambda}{2\pi\epsilon_1 a} = \frac{\lambda}{2\pi\epsilon_2 b} = \frac{\lambda}{2\pi\epsilon_3 c}.$$

Therefore,

$$\boxed{\epsilon_1 a = \epsilon_2 b = \epsilon_3 c}$$

This is the condition of capacitance grading or dielectric grading for equal maximum electric stress.

10.5 Voltage Expression

Total voltage between conductor and sheath is

$$V = \int_a^b E_1(r) dr + \int_b^c E_2(r) dr + \int_c^d E_3(r) dr.$$

So,

$$V = \frac{\lambda}{2\pi\epsilon_1} \ln \frac{b}{a} + \frac{\lambda}{2\pi\epsilon_2} \ln \frac{c}{b} + \frac{\lambda}{2\pi\epsilon_3} \ln \frac{d}{c}.$$

Therefore,

$$V = \frac{\lambda}{2\pi} \left[\frac{1}{\epsilon_1} \ln \frac{b}{a} + \frac{1}{\epsilon_2} \ln \frac{c}{b} + \frac{1}{\epsilon_3} \ln \frac{d}{c} \right]$$

Maximum electric field will not occur just off the inner conductor if another layer has greater maximum stress, i.e.

$$\epsilon_1 a > \epsilon_2 b \quad \text{or} \quad \epsilon_1 a > \epsilon_3 c$$

For equalized maximum stresses:

$$\epsilon_1 a = \epsilon_2 b = \epsilon_3 c$$

11 Question 4(b): Uniqueness Theorem

State and prove Uniqueness Theorem. Explain whether getting different solutions of Laplace's equation for a given problem by use of different methods is violation of Uniqueness Theorem or not.

11.1 Statement of Uniqueness Theorem

For a given region, if charge distribution and boundary conditions are specified, the solution of Poisson's equation is unique.

For charge-free region, if boundary conditions are specified, the solution of Laplace's equation is unique.

That means there cannot be two different correct potential functions satisfying the same differential equation and same boundary conditions.

11.2 Proof

Suppose two solutions exist:

$$V_1$$

and

$$V_2.$$

Both satisfy Laplace's equation:

$$\nabla^2 V_1 = 0$$

and

$$\nabla^2 V_2 = 0.$$

Both satisfy same boundary conditions.

Define

$$V_d = V_1 - V_2.$$

Then

$$\nabla^2 V_d = \nabla^2 V_1 - \nabla^2 V_2.$$

So,

$$\nabla^2 V_d = 0.$$

On the boundary, since both potentials are same,

$$V_d = 0.$$

Now use Green's first identity:

$$\iiint_V [|\nabla V_d|^2 + V_d \nabla^2 V_d] dv = \oint_S V_d \frac{\partial V_d}{\partial n} dS.$$

Since

$$\nabla^2 V_d = 0$$

and on boundary

$$V_d = 0,$$

we get

$$\iiint_V |\nabla V_d|^2 dv = 0.$$

Since the integrand is always non-negative,

$$|\nabla V_d|^2 = 0.$$

So,

$$\nabla V_d = 0.$$

Therefore V_d is constant.

But on boundary,

$$V_d = 0.$$

Hence,

$$V_d = 0$$

everywhere.

Thus,

$$V_1 = V_2.$$

Therefore the solution is unique.

11.3 Different Methods and Different Answers

If the same Laplace problem has same boundary conditions, different correct methods cannot give truly different solutions.

If two methods give different answers, then at least one of the following must be true:

- one solution has calculation error,
- one solution does not satisfy Laplace equation,
- one solution does not satisfy boundary conditions,

- approximate numerical solution has approximation error,
- the two answers look different but are mathematically equivalent.

So getting different exact solutions for same problem violates the uniqueness theorem.

But getting different approximate numerical values due to approximation does not violate the theorem if the exact solution is still unique.

For fixed boundary conditions, solution of Laplace's equation is unique.

Different correct methods must lead to the same solution.

12 Question 5(a): Capacitance Between Two Infinitely Long Transmission Line Conductors

Derive the expression of capacitance between two infinitely long transmission line conductors.

12.1 System

Consider two parallel infinitely long cylindrical conductors.

Each conductor has radius a .

The distance between their centres is D .

Let charge per unit length on one conductor be

$$+\lambda.$$

Let charge per unit length on the other conductor be

$$-\lambda.$$

The medium permittivity is

$$\epsilon.$$

We need capacitance per unit length.

12.2 Electric Potential Difference

For two parallel conductors, the exact potential difference between them is

$$V = \frac{\lambda}{\pi\epsilon} \cosh^{-1} \left(\frac{D}{2a} \right)$$

Using the identity

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}),$$

we can write

$$V = \frac{\lambda}{\pi\epsilon} \ln \left[\frac{D + \sqrt{D^2 - 4a^2}}{2a} \right].$$

12.3 Capacitance Per Unit Length

Capacitance per unit length is

$$C' = \frac{\lambda}{V}.$$

Substitute value of V :

$$C' = \frac{\lambda}{\frac{\lambda}{\pi\epsilon} \cosh^{-1}\left(\frac{D}{2a}\right)}.$$

Thus,

$$C' = \frac{\pi\epsilon}{\cosh^{-1}\left(\frac{D}{2a}\right)}$$

or,

$$C' = \frac{\pi\epsilon}{\ln\left[\frac{D + \sqrt{D^2 - 4a^2}}{2a}\right]}$$

For $D \gg a$,

$$\sqrt{D^2 - 4a^2} \approx D.$$

So,

$$\frac{D + \sqrt{D^2 - 4a^2}}{2a} \approx \frac{D + D}{2a} = \frac{D}{a}.$$

Therefore approximate formula is

$$C' \approx \frac{\pi\epsilon}{\ln(D/a)}$$

12.4 Exam Remark

This is capacitance per unit length.

For length l ,

$$C = C'l.$$

$$C' = \frac{\pi\epsilon}{\cosh^{-1}\left(\frac{D}{2a}\right)} = \frac{\pi\epsilon}{\ln\left[\frac{D + \sqrt{D^2 - 4a^2}}{2a}\right]}$$

13 Question 5(b): Field Components Above Earth Plane Due to Real and Image Charge

From the concept of electric dipole, derive the expressions for the components of electric field intensity at a given point above the earth surface due to a real point charge and its image charge.

13.1 Physical System

Let the earth surface be a grounded conducting plane.

Take this plane as

$$z = 0.$$

Let a real charge $+Q$ be placed at height h above the plane.

So the real charge is at

$$(0, 0, h).$$

By method of images, the grounded plane can be replaced by an image charge

$$-Q$$

placed at

$$(0, 0, -h).$$

The region of interest is above the plane, i.e.

$$z > 0.$$

This real charge and image charge form a dipole-like system.

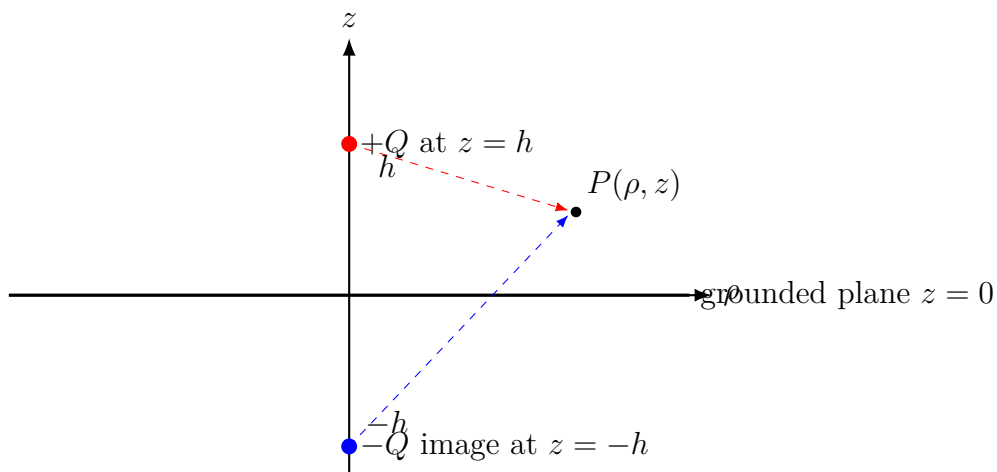


Figure 4: Real charge and image charge above grounded earth plane.

13.2 Distances

Let point of observation be

$$P(\rho, z).$$

Distance from real charge is

$$R_1 = \sqrt{\rho^2 + (z - h)^2}.$$

Distance from image charge is

$$R_2 = \sqrt{\rho^2 + (z + h)^2}.$$

13.3 Potential at Point P

Potential due to real charge is

$$V_1 = \frac{Q}{4\pi\epsilon R_1}.$$

Potential due to image charge is

$$V_2 = -\frac{Q}{4\pi\epsilon R_2}.$$

Therefore total potential above the plane is

$$V = \frac{Q}{4\pi\epsilon} \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

That is,

$$V = \frac{Q}{4\pi\epsilon} \left[\frac{1}{\sqrt{\rho^2 + (z - h)^2}} - \frac{1}{\sqrt{\rho^2 + (z + h)^2}} \right]$$

13.4 Electric Field Components

Electric field is

$$\mathbf{E} = -\nabla V.$$

In cylindrical coordinates,

$$\mathbf{E} = E_\rho \hat{\rho} + E_z \hat{\mathbf{z}}.$$

Also,

$$E_\rho = -\frac{\partial V}{\partial \rho}$$

and

$$E_z = -\frac{\partial V}{\partial z}.$$

Using direct field contribution of real and image charge,

$$E_\rho = \frac{Q}{4\pi\epsilon} \left[\frac{\rho}{R_1^3} - \frac{\rho}{R_2^3} \right]$$

where

$$R_1 = \sqrt{\rho^2 + (z - h)^2}$$

and

$$R_2 = \sqrt{\rho^2 + (z + h)^2}.$$

Therefore,

$$E_\rho = \frac{Q\rho}{4\pi\epsilon} \left[\frac{1}{(\rho^2 + (z - h)^2)^{3/2}} - \frac{1}{(\rho^2 + (z + h)^2)^{3/2}} \right]$$

Similarly,

$$E_z = \frac{Q}{4\pi\epsilon} \left[\frac{z - h}{R_1^3} - \frac{z + h}{R_2^3} \right]$$

So,

$$E_z = \frac{Q}{4\pi\epsilon} \left[\frac{z - h}{(\rho^2 + (z - h)^2)^{3/2}} - \frac{z + h}{(\rho^2 + (z + h)^2)^{3/2}} \right]$$

13.5 Check on Grounded Plane

On the grounded plane,

$$z = 0.$$

Then

$$R_1 = R_2 = \sqrt{\rho^2 + h^2}.$$

So,

$$E_\rho = 0.$$

This means tangential electric field on conducting plane is zero.

Normal component becomes

$$E_z = \frac{Q}{4\pi\epsilon} \left[\frac{-h}{(\rho^2 + h^2)^{3/2}} - \frac{h}{(\rho^2 + h^2)^{3/2}} \right].$$

$$E_z = -\frac{Qh}{2\pi\epsilon(\rho^2 + h^2)^{3/2}}.$$

This is normal to the conducting plane, as expected.

$$E_\rho = \frac{Q\rho}{4\pi\epsilon} \left[\frac{1}{(\rho^2 + (z - h)^2)^{3/2}} - \frac{1}{(\rho^2 + (z + h)^2)^{3/2}} \right]$$

$$E_z = \frac{Q}{4\pi\epsilon} \left[\frac{z - h}{(\rho^2 + (z - h)^2)^{3/2}} - \frac{z + h}{(\rho^2 + (z + h)^2)^{3/2}} \right]$$

14 Final Formula Sheet

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} \mathbf{R}$$

$$\nabla \cdot \mathbf{D} = \rho_v$$

$$\mathbf{E} = -\nabla V$$

$$E_{1t} = E_{2t}$$

$$w_e = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon E^2$$

$$p = \frac{\rho_s^2}{2\epsilon} = \frac{1}{2} \epsilon E^2$$

$$E_i(r) = \frac{\lambda}{2\pi\epsilon_i r}$$

$$V = \frac{\lambda}{2\pi} \left[\frac{1}{\epsilon_1} \ln \frac{b}{a} + \frac{1}{\epsilon_2} \ln \frac{c}{b} + \frac{1}{\epsilon_3} \ln \frac{d}{c} \right]$$

$$C' = \frac{\pi\epsilon}{\cosh^{-1} \left(\frac{D}{2a} \right)}$$

$$V = \frac{Q}{4\pi\epsilon} \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$